# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 6 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 5: **Prime Time** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The smallest digit to put in the blank to make 763_ divisible by 3 is: 1

**(a)** 2 **(b)** 4 **(c)** 3 **(d)** 6

**2.** The largest 3-digit multiple of 13 is: 1

**(a)** 999 **(b)** 962 **(c)** 988 **(d)** 975

**3.** How many prime numbers are there between 50 and 90? 1

**(a)** 8 **(b)** 11 **(c)** 9 **(d)** 10

**4.** The sum of all distinct prime factors of 330 is: 1

**(a)** 18 **(b)** 21 **(c)** 24 **(d)** 330

**5.** The number of factors of 72 is: 1

**(a)** 14 **(b)** 5 **(c)** 6 **(d)** 12

**6.** The HCF of 56 and 84 is: 1

**(a)** 168 **(b)** 56 **(c)** 28 **(d)** 84

**7.** The LCM of 15 and 25 is: 1

**(a)** 37 **(b)** 150 **(c)** 75 **(d)** 375

**8.** The smallest digit to put in the blank to make 819_ divisible by 3 is: 1

**(a)** 4 **(b)** 1 **(c)** 2 **(d)** 0

**9.** The largest 3-digit multiple of 17 is: 1

**(a)** 969 **(b)** 999 **(c)** 952 **(d)** 986

**10.** How many prime numbers are there between 30 and 80? 1

**(a)** 12 **(b)** 11 **(c)** 14 **(d)** 13

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Find the smallest number that must be added to 5678 to make it divisible by 11. 2

**12.** Find the missing digit so that 784_ is divisible by 9. 2

**13.** Find the HCF and LCM of 42 and 56 using prime factorisation. 2

**14.** Find the smallest number that must be added to 3456 to make it divisible by 7. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Find the prime factorisation of 252 and the number of its factors. 3

**16.** Two bells ring every 20 minutes and every 30 minutes. They ring together at 9 am. When do they next ring together? 3

**17.** Two ropes of length 90 m and 126 m are cut into pieces of equal length, as long as possible, with nothing left over. Find the length of each piece and the total number of pieces. 3

OR

Find the prime factorisation of 300 and the number of its factors. 3

**18.** Two bells ring every 24 minutes and every 36 minutes. They ring together at 9 am. When do they next ring together? 3

OR

Two ropes of length 56 m and 98 m are cut into pieces of equal length, as long as possible, with nothing left over. Find the length of each piece and the total number of pieces. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** A room is 600 cm long and 480 cm wide. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

OR

Find the smallest number that leaves a remainder 1 when divided by 5, 6 and 10, and the next such number. 5

**20.** A room is 390 cm long and 330 cm wide. Find the side of the largest square tile that fits exactly, and the number of tiles. 5

OR

Find the smallest number that leaves a remainder 1 when divided by 6, 8 and 9, and the next such number. 5

## END OF QUESTION PAPER — Total: 40 Marks

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